PRACTICAL 4

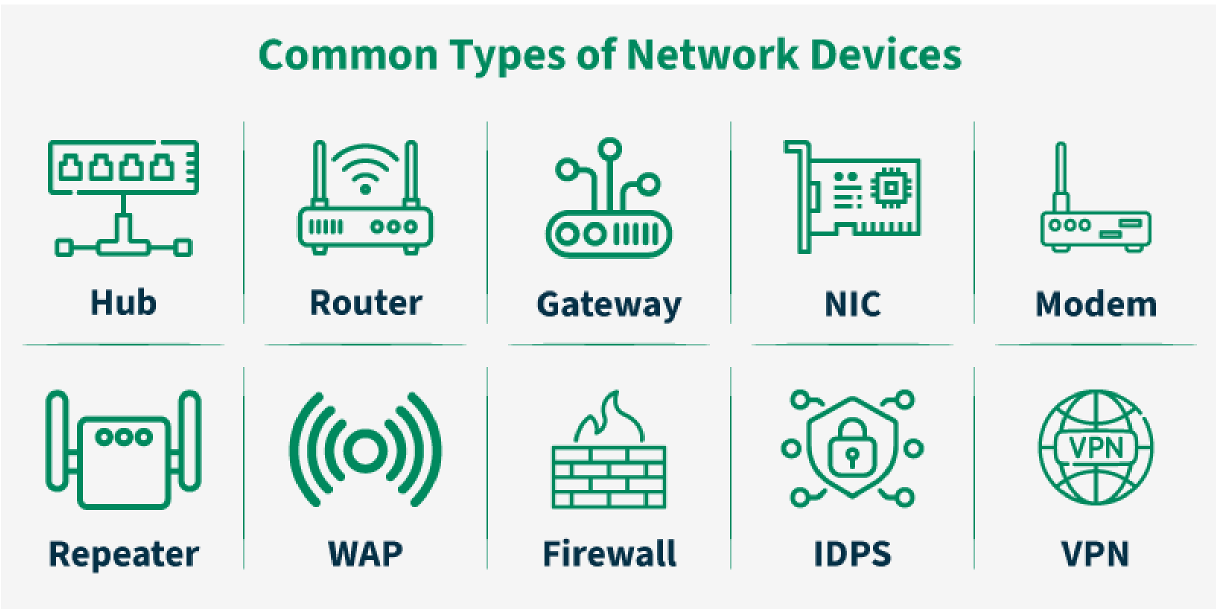
Aim : Study and Test various Network devices available at Department/Institute. (Repeater, Hub, Switch, Bridge, Router and Gateway).

Input :

Network Devices

Network Devices are the physical appliances required for communication and interaction between computers on a computer network.

- Enable communication by transmitting and receiving data between devices.
- Allow devices to connect to networks efficiently and securely.
- Improve network performance by reducing congestion and managing traffic.
- Extend network coverage and solve signal loss or attenuation problems.

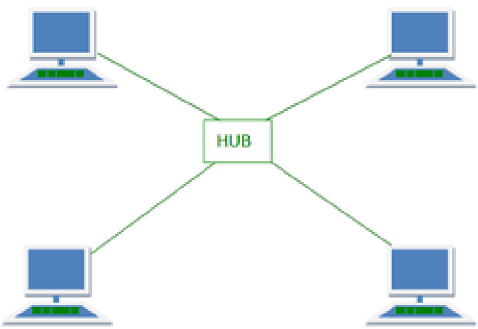


Types of Network Devices

Layer 1 Devices : Physical Layer

These devices deal with raw electrical or optical signals (bits). These devices don't understand IP addresses or MAC addresses.

1. **Hub**
   It is a central connection point for devices in a LAN.
   - It takes an incoming signal on one port and blindly broadcasts it to all other ports without knowing the recipient.
   - High network traffic collisions and security risks (everyone sees everyone's data).
   - It is a obsolete device, replaced by Switches.



Advantages

- **Low cost :** Hubs are very cheap compared to network switches.
- **Easy to use :** They need no setup or special software.
- **Media flexibility :** They can connect different cable types like twisted pair or fiber optic.

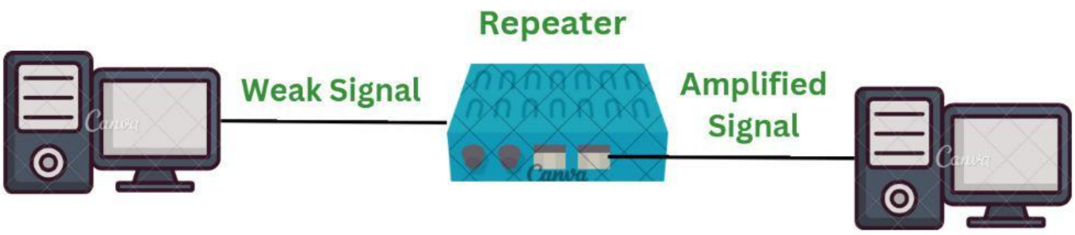
Disadvantages

- **High traffic :** They broadcast data to all ports instead of the target device.
- **Shared speed :** Total bandwidth is shared among all connected devices.
- **Low security :** Any connected device can see all network data.
- **Half-duplex only :** Devices cannot send and receive data at the same time.

2. **Repeater**

It regenerates signals to extend the range of a network.

- It receives a weak signal (due to attenuation over long cables), amplifies it, and retransmits it.
- Extending WiFi range or Ethernet cables beyond 100 meters.



Advantages

- **Extends range :** Overcomes signal loss (attenuation) over long distances.
- **Cost-effective :** Inexpensive compared to routers or switches.
- **Easy setup :** Simple to install and requires no complex software configuration.
- **No processing overhead :** Works transparently on raw bits without slowing down data via heavy processing.

Disadvantages

- **Amplifies noise :** Boosts background noise right along with the actual signal.
- **No traffic filtering :** Cannot manage, route, or filter data packets.
- **Shared bandwidth :** Does not segment network traffic, which leaves all connected devices in a single collision domain.
- **Limited quantity :** Too many repeaters added in a row can cause signal delays and packet collisions.

Layer 2 Devices: Data Link Layer

These devices work with MAC Addresses (Physical Addresses). They are smarter than hubs.

3. **Switch**
   A switch is a high-speed networking device that connects devices (computers, printers, servers) within a Local Area Network (LAN),
   - Unlike hub, switch learns the MAC address of every connected device.
   - It sends data only to the specific port where the destination device is connected.
   - Reduces collisions and improves security and speed.

Advantages

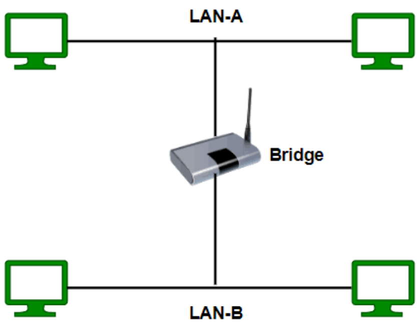
- **More Speed** : Each port gets full, dedicated bandwidth. Devices do not share speed.
- **Fewer Collisions** : Direct paths mean fewer data crashes on the network.
- **Better Security** : Data goes only to the intended device. It is harder to snoop.
- **Grows Easily** : You can add more ports and devices as needed.

Disadvantages

- **Higher Cost** : Switches cost more than simple hubs.
- **Setup Needed** : Managed types need a person who knows network rules to set them up.
- **Hard to Trace** : Finding a bad link or fault takes more time in big setups.
- **Limited Traffic Control** : They do not handle different subnets or internet routing like routers do.

4. **Bridge**
This connects two separate LAN segments to make them appear as one.

- Filters traffic based on MAC addresses to keep local traffic local, reducing congestion.
- Largely replaced by Switches (which are essentially multi-port bridges).



Advantages

- **Reduces traffic :** Breaks big networks into smaller parts to stop extra data flow.
- **Fewer collisions :** Creates separate collision domains to give devices faster speeds.
- **Easy setup :** Simple to install without extra complex software.
- **Extends network :** Acts like a repeater to make the network longer.

Disadvantages

- **Slower than switches :** Slower at processing data than modern network switches.
- **No high security :** Does not have advanced security tools like firewalls.
- **Broadcast issues :** Can pass broadcast messages through, which may slow down performance.
- **Limited use :** Works only on local area networks (LANs), not wide area networks (WANs).

Layer 3 Devices: Network Layer

These devices work with IP Addresses (Logical Addresses) and connect different networks together.

5. **Router**
   It is a networking device that connects multiple networks and directs data packets between them using IP addresses.
   - It uses IP Addresses to determine the best path for data packets to travel.
   - It maintains a Routing Table to make these decisions.
   - Routers stop broadcast traffic, isolating networks from each other.

Advantages

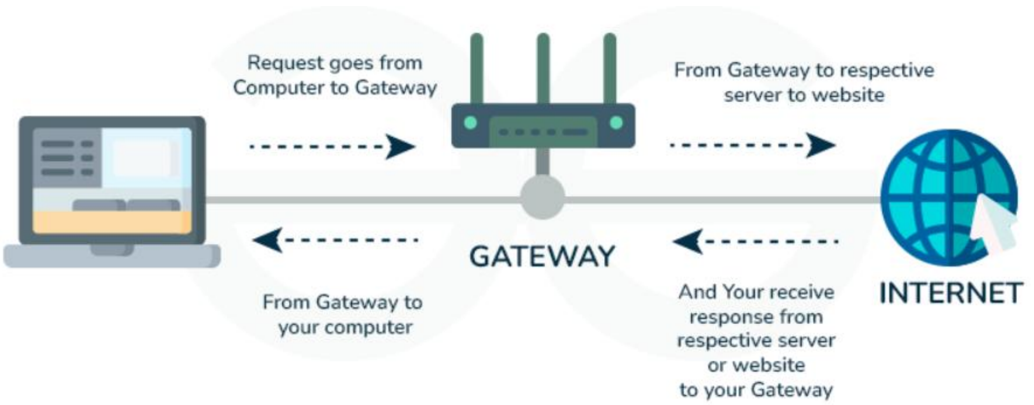
- **Internet Sharing :** Allows many devices to share one internet connection.
- **Path Selection :** Finds the fastest and best route for data packets.
- **Network Security :** Protects internal devices using Network Address Translation (NAT) and firewalls.
- **Traffic Control :** Manages and separates network traffic to reduce congestion.

Disadvantages

- **Higher Cost :** Advanced or enterprise models cost more than basic switches or hubs.
- **Range Limits :** Wireless signals weaken over distance and through physical walls.
- **Setup Complexity :** Configuration requires basic technical knowledge.
- **Single Point of Failure :** If the main router stops working, connected devices lose internet access.

Layer 4-7 Devices: Advanced Processing

6. **Gateway**
   It is a hardware device or software node that acts as an entry/exit point between two distinct networks using different protocols, translating data to ensure compatibility.
   - It can convert protocols, data formats, or architectures (e.g., connecting a TCP/IP network to a legacy Mainframe network).
   - An Internet Gateway translates your private LAN requests into public internet requests



Advantages

- **Protocol conversion :** Translates data from one network architecture to another.
- **Network security :** Filters traffic and blocks external threats.
- **Wide connectivity :** Links completely different systems like LANs to WANs or the internet.
- **Centralized control :** Manages data routing efficiently.

Disadvantages

- **High cost :** Expensive to install and maintain.
- **Complex configuration :** Hard to set up and manage.
- **Slower performance :** Protocol translation can add processing delays.
- **Single point of failure :** If the gateway fails, connected networks lose communication.